INTERNET OF THINGS GROUP ASSIGMENT

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SMART AGRICULTURE IN KENYA : SAFARICOM DIGIFARM CASE STUDY

Quick Summary

This report explores how the Internet of Things (IoT) is being used in Kenyan farming, using Safaricom's DigiFarm as a prime example. It looks at how connecting devices, farmers, and services can help farms be more productive and make smarter choices. The report looks at: the IoT food chain worldwide, big tech companies that dominate the tech, a plan for a Kenya-friendly IoT business, and ways Kenya can get more from IoT through policies and teamwork.

The study suggests that while DigiFarm relies on global giants like Amazon and Microsoft, things could be better if it invested in local data centers, backed open-source tools, and made sure farmers owned their data. A mix of subscription and pay-as-you-go plans could give farmers access to cheap IoT services and boost local startups.

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1. Introduction

Agriculture is still important to Kenya, employing most people in rural areas and making up a big chunk of the country's money. With problems like climate change and tricky markets, IoT tech can help farmers be smarter and more efficient. IoT links things like sensors, apps, and drones to systems that collect and analyze info in real-time.

In Kenya, companies like Safaricom are using IoT to connect small farmers to digital markets. DigiFarm, which started in 2017, uses IoT with mobile money and advice to help farmers grow better crops and manage their farms better.

2. Case Selection and Context

Safaricom DigiFarm is a platform that helps small farmers. It connects them to banks, suppliers, and buyers using phones. Farmers can get weather updates, soil info, and training through apps.

Key Points:

- Started:2017
- Partners: iProcure, FarmDrive, Arifu, and KCB Bank
- Farmers Using It:Over 1.4 million by 2023
- Goal: Help small farmers grow more and earn more using digital tools.

DigiFarm uses IoT sensors on farms to track soil, weather, and fertilizer. Farmers get this information on their phones, which helps them make better decisions. This also helps organizations keep track of food supplies and plan deliveries.

DigiFarm is local, but it depends on cloud services from other countries, foreign gadgets, and expertise from overseas. This means it relies a bit on global companies.

3. Mapping the Global IOT Value Chain

3.1 Components and flow

An IoT agricultural system like DigiFarm consists of:

1. Devices: Sensors for soil, temperature and humidity. Weather stations collect these data. The devices are mostly imported.

2. Connectivity: Safaricom's network connects the devices. In remote areas, satellite or Wi-Fi is used.

3. Platforms: These gather data, analyze it, and display it for farmers and adminstrators.

4. Applications: Apps give farmers information in a simple way.

5. Services: These include advice, credit, training, and market connections.

FLOW: Sensors → Connectivity → Cloud Platforms → Data Analysis → Apps → Farmer Services.

3.2 Local vs International control

Component	Controlled Locally (Kenya)	Controlled Internationally
IoT Devices	Installation, field support, and repairs by local technicians	Device manufacturing mainly in China and Europe
Connectivity	Safaricom provides network services and local SIM-based IoT connectivity	Core infrastructure from Huawei, Cisco, and Ericsson
Platforms	Local management systems and payment integrations handled by Safaricom	Cloud hosting, backups, and analytics done via AWS or Azure
Applications	Designed locally with Swahili language options and local developer input	Some advanced analytics tools integrated from Google or IBM
Services	Training, support, and financial services managed by Safaricom and partners	Some AI-based forecasting tools developed by global firms

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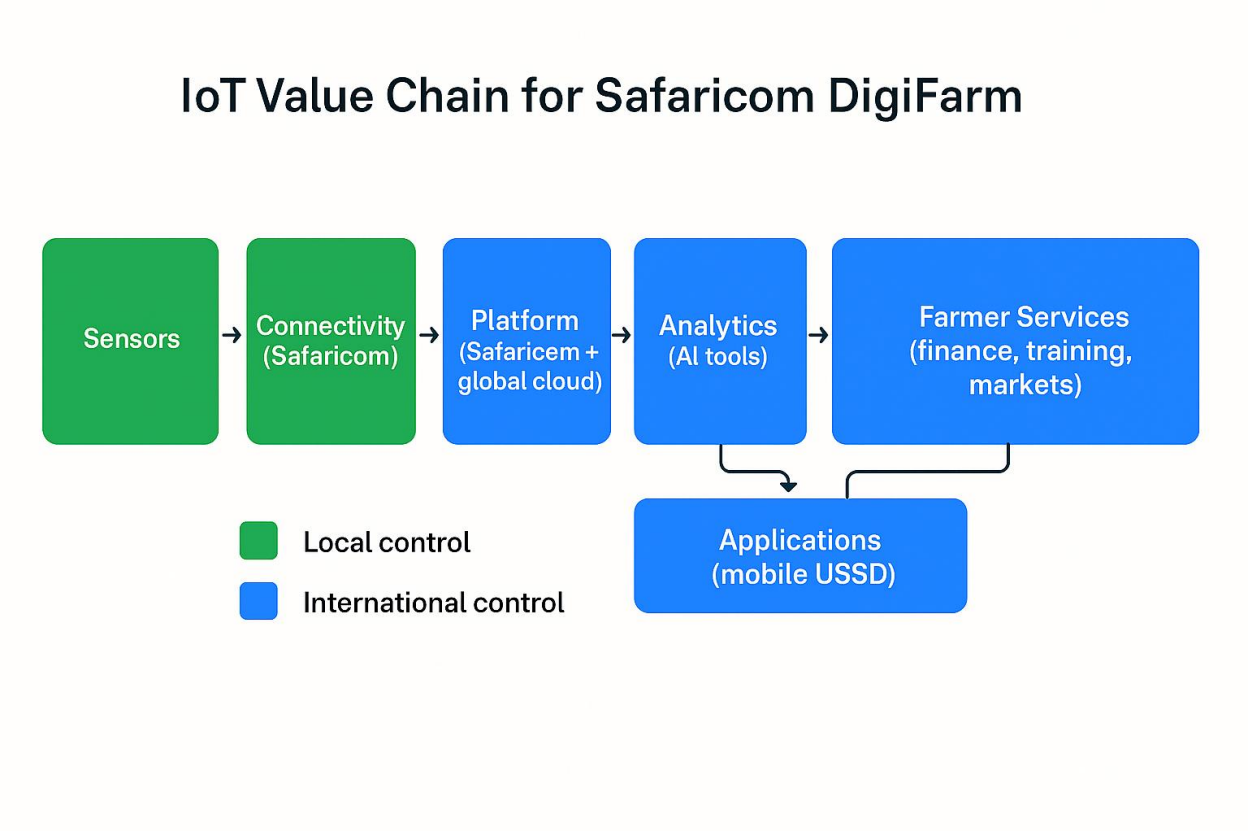
3.3 Diagram of IOT value chain



4. Role of Global Information Monopolies

4.1 Who's Involved

- Amazon Web Services (AWS): Stores data and manages IoT systems.
- Microsoft Azure: Analyzes data and provides AI tools for data visualization.
- Google Cloud: Machine learning tools for predictions.
- Huawei: Provides network equipment and IOT connectivity hardware.

These companies are giants in IoT, so Kenyan projects depend on them for hosting, data processing and connectivity.

4.2 Risks for Kenyan Farmers and Businesses

1. Data Control: Farmers' data is stored abroad, meaning less control.
2. Stuck with One Provider: Switching can be hard and costly.
3. Price Changes: Exchange rates can push up prices for users.
4. Few Local Experts: Reliance on outsiders slows local progress.
5. Cybersecurity Risks: Data on foreign servers could get hacked.

4.3 Opportunities for Local Players

1. Local Data Centers: Safaricom and others could build local cloud capacity.
2. Edge Computing: Process data locally to cut costs.
3. Training and Jobs: Universities can train students in IoT and cloud tech.
4. Teamwork: Startups and large companies can work together on local solutions.

5. Sustainable IOT Business Model for Kenya

5.1 Overview

A Kenyan startup can create IoT services to add to DigiFarm like smart irrigation, pest control, and crop tracking using local servers and apps. This helps farmers get affordable tools without relying too much on global providers.

5.2 Chosen Revenue Model: Hybrid Subscription + Pay-per-Use

- Subscription: Monthly access to advice and sensor data.
- Pay-per-Use: Pay only for advanced services like drone monitoring.
- Free: Basic weather and soil updates are free.

5.3 How It Lessens Dependence

1. Use local hosting and open-source platforms.
2. Partner with Kenyan universities and hubs for innovation.
3. Create local data ownership through farmer groups.
4. Train young people to fix devices and write code.

5.4 Example of Implementation

A startup like AgriSense Kenya could create solar-powered IoT sensors that send farm data to a local platform. Farmers get alerts via SMS, and the data helps predict drought or disease risks. The company could work with Safaricom for connectivity and with local governments to reach farmers.

6. Critical Reflection: Policies, Partnerships, Innovations

6.1 Policies

- Data Protection: Store sensitive farm data locally.
- Tax Incentives: Help Kenyan startups build IoT infrastructure.
- Grants: Government and NGOs can fund local IoT projects.
- Open Standards: Ensure systems can integrate easily to avoid depending on one.

6.2 Partnerships

- Government + Private Sector + Universities = shared research.
- Farmers' cooperatives to manage and own data.
- International teamwork focusing on sharing technology.

6.3 Innovations

- Edge Analytics: Process data near farms.
- Local Hardware: Assemble sensors locally.
- Data Cooperatives: Farmers can profit from data use.
- Community IoT Hubs: Provide training and shared devices for small farmers.

6.4 How Power Can Shift to Local Innovation Ecosystems

1. Local Infrastructure: Get more data centers and IoT labs built in Kenya.
2. Government Support: Prioritize Kenyan tech firms on IoT government contracts.
3. Open Source Tools: Use open source tools so we aren't stuck with just one company's tech.
4. Public Funding: Give grants to local researchers and new businesses.
5. Education and Training: Start IoT courses in colleges.
6. Inclusive Innovation: Encourage young folks and ladies to get into IoT.
7. Regional Collaboration: Work with other African countries to build common digital spaces.

These steps would help Kenya depend less on foreign providers.

7. Conclusion

Safaricom DigiFarm shows how IoT can help Kenyan farmers make smart decisions, grow more, and connect to markets. Kenya can change this by supporting technologies. The future of IoT in Kenyan agriculture depends on supporting local startups and farmers to lead innovation.

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